AI-DRIVEN PLATFORM GUIDING PADDY FARMERS ACROSS THE FULL CULTIVATION LIFECYCLE

Project ID: 25-26J-242

Project Proposal Report

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Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

AI-DRIVEN WEED DETECTION IDENTIFICATION AND STAGES IDENTIFICATION

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DECLARATION

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I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

.....

Signature of the supervisor:

31/08/2025

Date:

ABSTRACT

Paddy is a major component in the agriculture of Sri Lanka and inefficiency in the manner in which the field is tended leads to less yields, and high cost of production. Generalized control methods of weed management like manual clearing and universal application of herbicides have usually been inefficient, not environment-friendly and economically costly to the smallholder farmers. To overcome this obstacle, the proposed research suggests developing an artificial intelligence-based mobile platform that implements computer vision, deep learning, and expert knowledge that can be local and with which paddy farmers can be assisted throughout the entire cultivation process. It uses convolutional neural networks (CNNs) to recognize the widely prevalent weeds in Sri Lanka, even when they grow to and match paddy in size. It also categorizes the stages of weed growth, the degree to which an infestation exists and the recommendation on the proportionate use of herbicide based on herbicides locally available, nature of dose recommendation and conditions surrounding the application. Moreover, causal reasoning against the factors like irrigation and soil management is added along with proactive measures of how to avoid such recurrence. Focused on offline use, multi-language capabilities and with a user-friendly interface that supports farmers, the solution guarantees the accessibility of the product in the rural areas, where digital literacy rates are low. The proposed platform will facilitate sustainable farming because it will support sustainable farming practices, as it reduces the cost of herbicide misuse and improves crop yields, in addition to enabling farmers to get real-time actionable guidance. The study therefore will fill important gaps in the management of paddy weeds in tropical regions with the best combination of new AI solutions with local area solutions to enhance food security and agricultural vulnerabilities in Sri Lanka.

Keywords: Artificial Intelligence, Convolutional Neural Networks (CNN), Weed Detection, Growth Stage Classification, Paddy Cultivation, Mobile Agriculture Application, Sustainable Farming, Decision Support System, Sri Lanka

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LIST OF ABBREVIATIONS

Abbreviation	**Description**
AI	Artificial Intelligence
CNN	Convolutional Neural Network
ML	Machine Learning
API	Application Programming Interface
RRDI	Rice Research and Development Institute

1 INTRODUCTION

1.1 Background & Literature survey

1.1.1 Background

Agriculture is the pillar of the Sri Lanka economy and one of the economic most popular and economically important agricultural activities within the Island nation is the paddy cultivation [1]. But modern paddy production has never been in a more perilous situation as it is today with challenges to productivity and sustainability. Of these, the invasion of weeds has proved to be one of the most raging and economically destructive problems to the farmers across Sri Lanka [2]. Weed control in paddy growing can be impounding and goes beyond mere identification of weed plants. Conventional weed management practices have largely been based on a manual method of identifying the weed, timing of herbicide phenomena on universal schedules and decision making on the basis of farmer experience [3]. Such traditional methods are capable of failing in managing the varied and changing weed problems typical with systems of modern agricultural systems. Past research has revealed that weed competition may decrease mean paddy crop by 20-40 percent, with competition during early season especially harmful to crop establishment and further maturation [4]. Economics Inefficient weed control has more consequences than the direct loss of yield, including a rise in the cost of inputs, labor, and long-term degradation of soil health. Moreover, misuse of herbicides such as the application practices leads to environmental issues and evolution of population resistant to the herbicides [5].

The challenges faced by farmers in identifying a weed are considered complex and especially in early stages of growth when the weed is easier to address after being identified. Lots of species of weeds bear a striking similarity with paddy seedlings at early stages of growth, thus are hard to tell apart unless one has specialized knowledge [6]. The difficulty in identification is further enhanced by the varied

and complex ecology of weeds that occur in various agro-ecological regions of Sri Lanka, and where that is in the form of different types of soil, temperature control methods and different weather conditions [7].

The stage of weed growth is the key factor in defining control options efficiency especially selection and dose of herbicides. Plants at different growth phases will be sensitive to different chemical treatments, and early application will normally be more effective and less costly than later application [8]. Nevertheless, farmers have no convenient instruments to identify weeds within a given stage in time, which causes them to intervene too late and prompts them to use more herbicides, spending more money and causing environmental pollution. Enhanced weed management has implications not only to the productivity of individual persons in farms, but also the national goals as it relates to food security and rural development. Department of agriculture Sri Lanka has indicated that increased control of weeds as one of the priorities in the increase of Paddy production in the country and decrease of post-harvest losses [9]. Efficient rice production could be made by effective weed management technologies and could also support rural livelihoods. All these require smart, context-specific solutions that can give farmers the right, timely, and actionable information in their weed management. The development of advanced technologies in artificial intelligence, computer vision, and other areas opens new possibilities in the creation of technically sophisticated but accessible tools that will be able to meet the multifaceted needs of modern weed management and be usable by resource-limited smallholder farmers.

Weeds in paddy cultivation.

Grasses	Sedges	Broadleaved spp.
Echinochloa colona	*Cyperus iria*	*Commelina diffusa*
Echinochloa crus-galli	*Cyperus difformis*	*Eclipta alba*
Panicum repens	*Cyperus rotundus*	*Eichhornia crassipes*
Ischaemum rugosum	*Fimbristylis miliacea*	*Monochoria vaginalis*
Isachne globosa	*Linderina spp.*	*Murdannia nudiflora*
Fimbristylis dichotoma	*Schoenoplectus supinus*	*Sphenoclea zeylanica*
Paspalum distichum		*Ludwigia perennis*
		Haeranthus africanus

Figure 1.1: Sri Lankan paddy field weed species catalog

(source: Department of Agriculture Sri Lanka, 2025)

1.1.2 Literature Survey

Artificial Intelligence-Based Weed Detection Research Worldwide Studies

The recent developments in the field of artificial intelligence and computer vision led to a paradigm shift in the agricultural research field and many studies investigated automated weed detection systems. Deep learning methods, and specifically Convolutional Neural Networks (CNNs), have recently proven very successful in the classification of images applicable to agricultural contexts [10]. The primary tenets of AI-powered weed management have been laid out by international research, but the majority of the works have focused on crops grown under a temperate climate. One study has shown that under the theoretically flawless imaging conditions, CNN-based models are effective in discriminating between crop and weed in wheat and corn systems and the accuracy can reach over 90 percent [11]. Their study emphasized the possibility of democratising access to expert agricultural knowledge through AI applications based on mobile devices, especially where there could be limited extension services in developing countries.

On the same note, Detection of weeds in soybean fields as demonstrated by ResNet models is indeed feasible, especially when various training datasets are used [12].

CNN-Based Strategies to Agricultural Applications

The development of CNN architecture has achieved great advances in agricultural computer vision. MobileNet based architectures to detect weeds in real-time have been investigated and have achieved processing speeds compatible with smartphone implementations as well as overall detection rates of 85-percent or greater [13]. Those authors particularly talked about the computational restrictions that are commonly seen in the field of agriculture and proved the possibility that the usage of edge computing applied to precision agriculture. Further development of the study, Multi-stage classification systems have been designed to recognize weeds and the stage of development using morphological analysis [14]. Their method obtained an 87% accuracy in differentiating seedling, vegetative and flowering stages of five common weeds species which is directly pertinent to the delineation of the optimum time of treatment. This study sets some significant precedents in growth stage classification methodology which are used to make decisions on treatments.

The Classification of Weed Stages and Weathering Effects And Impact Assessment Research

Expert studies on the identification of the growth stage of weeds have become an important factor in precision agriculture. In a recent study Rodriguez et al. [15] optimized deep learning models to specifically classify the weed phenological stages by proving that the various stages of weed development all have different morphological features, which can be identified using computer vision. Their work demonstrated that efficacy of herbicides differ greatly by growth stage with early-stage intervention being 60-80% more effective than late-stage treatment. Thompson and Anderson damage assessment research has produced the methodologies to quantify the weed competition effect on crop yield based on the

image analysis [16]. Their models accurately forecasted yield losses with respect to weed density, weed species composition and growth stage distribution, which gives farmers economic rationale to adoption of treatment choices. This foundation of research becomes more applicable in coming up with decision support systems which can strike the balance between the cost of treatment and the yield gains.

Asian Rice Systems Development Initiatives

There have been few specific studies that have looked into the practice of weed management in Asian rice systems. A study by Sharma et al. [17] studied the application of AI-based weed detection to rice paddies in an Indian setting with specific challenges in flooded field regimes and tropical weeds. Their work reported 78% accuracy in identifying common Indian rice weeds, with large variation with the changes in the environment. The Chinese research done by Liu and Wang came up with mobile applications that identify rice weeds with consideration of farmer-friendly interface in that they needed to be able to work with no connectivity [18]. The inclusion of local herbicide databases and other cost optimization capabilities in their system supported the practical means of incorporating AI detection into treatment advice. Nevertheless, their specialization in Chinese species of weed and weed cultivation hinders their application in Sri Lankan conditions. A Philippine study by Santos et al. [19] evaluated economic benefit as weeds were controlled by AI-aided weed management in smallholder farms in rice-based cropping, where farmers were provided with accurate and timely identification and treatment recommendations, which the study found promising yield gains of 15-25 percent. They stressed the need to consider localized recommendations of herbicides and cost implications to farmers who are resource dependent.

Edge AI and Mobile computing in real-time agriculture

The merger between mobile technology and AI powered solutions in agriculture has gained a lot of traction as more villagers join the smartphone user base. According to research by Perera and Wickramasinghe [20], the mobile agricultural application has high potential of adoption in Sri Lanka, as long as the application is designed with the right user interface via the use of localised content. Their research named the major identifying success criteria with regard to influencing the farmer acceptance as offline capability, availability of the multi-language and the compatibility with the local farming conditions. Edge computing has also taken technological leaps forward which have made powerful AI processing possible on mobile devices. Kim et al. [21] have shown they were able to deploy TensorFlow Lite models to perform real-time monitoring of crops in a way that can feasibly be used in agricultural systems with processing achieving times of less than 3 seconds per image and a detection accuracy of agricultural interest. They also use optimization methods allowing high-complexity CNN models to run under resource-limited mobile hardware. Recent advances in the field of mobile-based AI applications featured an appealing application in the agricultural sector have been made further possible by the latest breakthroughs in model compression and quantization. This was demonstrated in a study by Brown et al. [22] using properly optimized models, 95 percent of full-precision accuracy can be obtained with 75 percent fewer computations, resulting in the possibility of real-time analysis even on entry-level smartphones accessible to the majority of farmers in developing nations.

Treatment Recommendations Decision Support Systems

Precise studies are starting to consider aspects of combining the ability to detect weeds with solution treatment suggestions. Martinez et al. [23] have devised comprehensive decision support systems with complex integration of weed identification and the combination of herbicide selection algorithms operating on various factors including the weed species, growth phase, weather/environment

constraints, and economic limitations. Their system had a farmer satisfaction of 92 percent and they proved to wipe off an average of 20 percent in herbicide expenses as a result of optimal treatment recommendations. Johnson and Clark [24] recommended that resistance management involving herbicides should be avoided through the intelligent treatment rotation and timing. Their AI-based suggestions were able to maintain herbicide efficacy on multi-year scales, with a 30% reduction in the total usage of chemicals consumed showcasing the possibility of having sustainable solutions of controlling weeds.

The Research Needs in Tropical Paddy Systems

In spite of the major progress in the research on agricultural AI, there has been a predominance of studies on the crops grown in temperate climates and weed species without much consideration to tropical paddy system widespread in South and Southeast Asia [25]. Sri Lankan agricultural paddy cultivation environments are unique in terms of the growing conditions, ecology of weeds and practices of agriculture and thereby necessitate special solutions keeping in view the local conditions and limitations. Strengths of current research. A primary factor that restricts current research is the lack of tropical weed species representation in data-sets, deficient knowledge of seasonal variances in the pattern of weed emergence in monsoon conditions and underrepresentation of local agricultural knowledge and practices are the key respects that limit current research. Not only that, economic and infrastructure limitations, which are prevalent in the system of smallholder farmers in the developing countries have not been fully taken into consideration in most current systems.

1.2 Research Gap

The modern day methods of controlling weeds in paddy cultivation majorly rely on manual weeding or general use of herbicides. These practices have been extensively deployed although the traditional practices are slow, consume a lot of labour and they are sometimes ineffective. Conventional forms of detecting weeds, which are mainly through visual observation by the farmer, are highly susceptible to human error during initial stages of weeds growth because they almost look similar to paddy. This either leads to untimely intervention or wrong treatment which spells losses in yields and wastage of money.

Despite the current impressive progress in the AI and computer vision in the agricultural industry, few of the current weed management technologies stand up to the requirements of the paddy farmer because several important gaps fragment the existing environment.

Deficiency of localized Solutions

Currently available weed detection systems have been developed in temperate climate crops and their performance in identifying the Sri Lankan paddy cultivation endemic weed species could not be predicted effectively. The tropical growing environment, varied weed ecology and particular farming conditions in Sri Lanka necessitate the use of specialist models that have been trained using local characteristics. Existing international solutions are not contextually specific to ensure happens upon implementation in south Asian agricultural systems.

Limited Growth Stage Integration

The majority of modern studies in detection of weeds are binomial (crop/weed) and exclude thorough analysis of the growth stage. This shortcoming does not allow best timing and dose suggestions of the herbicides, which contributes to ineffective herbicide use and unoptimal control results. Lack of growth stage consideration in current systems constitutes a big gap in the provision of actionable agricultural intelligence.

Lack of Integrated Treatment recommendation Guidance

Most of the focus of current researches has been on accuracy in detection of the fungus without including a blend of feasible treatment advisories that have resource availability using local herbicides, economic factors and farmer abilities. The difference between detection technology and implementation in the field restricts practical use of the existing.

Poor causal analysis Integration

Most of the systems already present do not tackle the underlying environmental and management causes of the emergence of the weeds. The limitation of the condition does not allow the development of multi-faceted field management models and lower possibilities of prophylactic measures in relation to weed treatment.

Lack of Rural Accessibility

Some of the potential AI solutions in agriculture include the use of high-end hardware, reliable internet connection, or complications in the usability model that cannot be implemented in rural Sri Lanka, and among poor farmers. The digital divide in the rural areas and the problem of infrastructure is still not adequately solved by the contemporary technological solutions.

Limited economic optimization

The current solutions do not take into consideration factors like the economic aspects including the cost of the herbicides, the ratio of treatment effectiveness and the amount of returns in terms of capital investment and also tend to ignore the interests of resource-limited smallholder farmers in Sri Lanka.

Table 1.2: Comparison of former research

Application Reference	Localized Weed Detection (SL)	Weed growth stage identification	Causal analysis (environmental & management factors)
Research[3]	No	No	No
Research[6]	No	No	No
Research [20]	No	No	No
Research[7]	Partial(weed)	No	Partial
Proposed System	Yes	Yes	Yes

This comparison table presents serious research gaps, as previous studies (Research [3], [6], and [20]) do not answer any of the three important constituents of localized weed detection, identification of growth stage, and causal analysis, and Research [7] only answered some of the aspects, with basic weed detection, and less causal analysis.

Solution is the first of its kind as it incorporates as have suggested, all the three key components, localized detection of weeds, especially in Sri Lankan paddy fields, identification of the stage of weed growth and the provision of causal analysis, including environmental and management-related factors. This is a tremendous improvement of current fragmented solutions and solutions and this offers the farmers a comprehensive, detailed and context-sensitive solution that goes beyond mere detection to provide actionable information dependent upon growth stages and underlying causal factors.

1.3 Research Problem

Weed infestation is an ongoing problem in Sri Lankan paddy cultivation, where uncontrolled prevalence not only competes with crops on nutrients, sunlight, and water but also brings a negative effect in yield level and increasing the cost involved. Generally people adopt manual or generalized application of chemical herbicides which leads to over-use of chemical and unwarranted expenditure as well as the destruction of soil in the long term. Our Ai-Powered Weed Detection and Growth Stage Identification module aims to alleviate these constraints by applying computer vision and deep learning algorithms to provide actionable insights on weed management that enable farmers to manage their crops on a real-time basis and exactly where they are needed by using mobile technology.

The system uses the images captured through mobile phones, where it applies the use of convolutional neural networks (CNNs) to differentiate between the paddy crop and the common weed species, even at their early growth stages. It also determines the growth condition of the weeds (e.g. seedling, vegetative, flowering) and the level of infestation, by which treatment can be optimized according to the growth stages. Combining the knowledge of herbicides that are available locally, the system proposes viable treatment, incorporating dosage information and application timing.

Along with analyzing root causes of emergence of weeds such as poor drainage or excess moisture, AI models can provide environmental and management strategies that can prevent or alleviate the emergence of weeds, providing both reactionary and proactive strategies that help tackle the problem. Farmer accessibility feature is what the system is aimed at. The suggested device has a mobile-friendly, easily navigable interface that makes even non-technical users benefit of the real-time, AI-guided suggestions. Not only does it minimize the reliance on too much herbicide, but it also promotes sustainable trends and economies.

The innovation is in integrating the detection of weeds, growth stage analysis, and localized treatment recommendation, with understanding root cause and end-user design. The platform is empowering farmers with actionable, real-time guidance to help them change the ways they deal with weeds: making weed control more accurate, efficient and sustainable, empowering farmers and making their land more productive.

2 OBJECTIVES

2.1 Main Objectives

The aim would be to design and deploy an AI-driven solution to predict precise identification of weeds, determination of their growth stages and provide a localized cost-effective treatment recommendation to the paddy farmers of Sri Lanka using the mobile-based platform. It will also analyse the severity of infestation and optimise the timing and dosage of treatment and also give an analysis of environmental and other management factors to advise preventive measures. In this way, it will reduce the misuse of herbicides, raise crop output, minimize production cost, and support long-term effects of sustainable farming.

2.2 Specific Objectives

The initial aim of the current research is to develop a complete detection and classification model of the weed species that suits the case of the paddy field in Sri Lanka. With modern convolutional neural networks (CNNs), the system will be learned using a labeled data set of local weed images taken at various levels of growth, during different times of day and in various conditions of environmental factors. The aim is to have a high accuracy of identification of at least fifteen large weeds with over 90% accuracy and real-time processing that is optimized to mobile platforms. This will enable the farmer to take field photographs and obtain quick and certain response without the high cost of equipments.

The second goal is dedicated to the determination of stages of weeds growth and its damage. The system will work by designing algorithms that can recognize the stages involved in the development of the plants such as the seedling stage, the vegetative stage, the flowering stage, and the seed production stage with a high rate of accuracy thus allowing interventions. A quantitative model of damage

evaluation will be too developed that will help assess the suspension of invasions of weeds and their competitive effects on crop production. The classification system will in-turn prioritize control measures so that treatment outcomes recommendations are both urgent and context-sensitive.

The third goal is a locally adapted treatment recommendation system with a combination of agronomic expertise and AI-generated expertise. This solution will offer local practical advice on herbicides that are locally available with the index of their effectiveness and the cost. It also will maximize calculation of dodge doses relying on development phases of weed, and severity in infestation, and also depending on environmental conditions. The recommendation engine will also enable the small-scale farmers to use the most ideal yet cost effective treatment by integrating cost optimization functionalities.

The last list of objectives is referring to causal analysis, preventive management, and farmer accessibility. Soil, irrigation, nutrient management and cropping practices will also be studied and I will develop models using machine learning to delve into the causes of the emergence of the weeds. The predictive features will also give a prediction of what risks might happen and give a farmer proactive solutions rather than only reactive ones. These features will be incorporated into a mobile app that is intended to be based on the principle of farmer-centricity such as multi-language accessibility (Sinhala, Tamil and English), offline functionality, and an easy user interface tailored to different levels of digital literacy. Combining novel AI capabilities with local usability ensures that the application will be highly adopted and translate to actual on the ground results by Sri Lankan paddy farmers.

3 METHODOLOGY

3.1 System Diagram

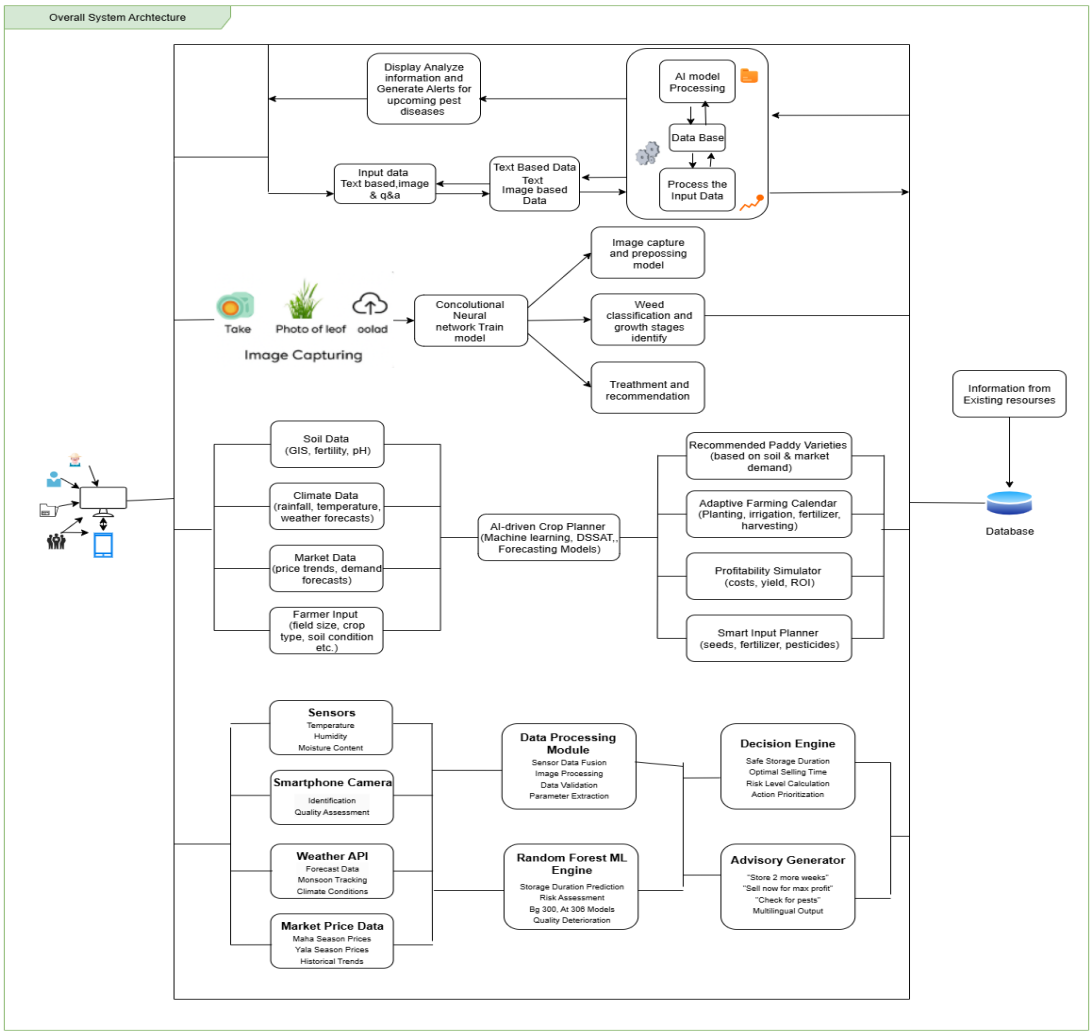

Figure 3.1: System Overall diagram

The system will be characterized by four dedicated AI-based modules created by the members of our team: a Crop Establishment Planner, which will welcome new farmers; an AI-based Weed Detection and Stage Identification component, which will identify weeds and crop growth phases with the help of image processing; a Pest Attack Forecasting system, which will send early warnings and make them

avoid it; and an AI-based Post-Harvest Advisor, which will optimize activity after harvest. These modules have a coordinated relationship, sharing data and insights via a centralized database and processing engine.

The platform is an actionable recommendations platform enabling farmers to receive optimal planting schedules, suggested paddy types in accordance with the soil and market conditions, intelligent encouragement of the application of fertilizers and pesticides, and simulated calculations of profits. This integrated solution can enable farmers to maximize the value of their paddy crop and reduce risks and maximize their investment using data-driven insights by combining real-time monitoring, predictive analytics, and expert advisory systems in the entire paddy cultivation process.

3.2 Component Diagram

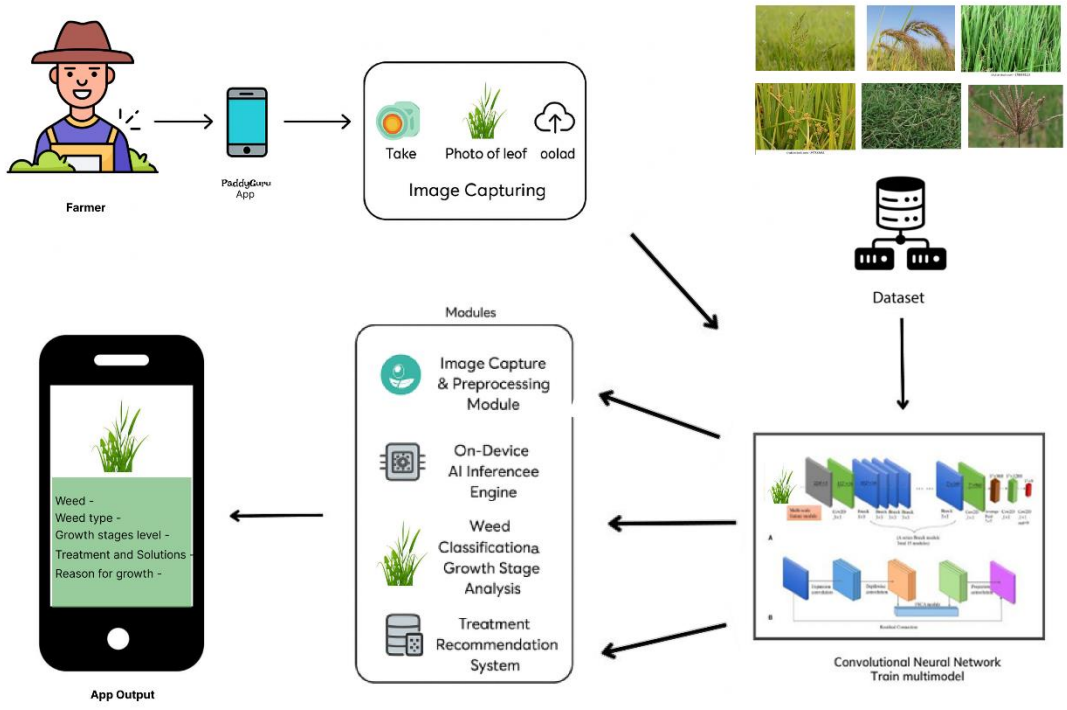

Figure 3.2:Component diagram

The proposed AI-based paddy weed detector may take the form of a hybrid mobile-cloud-based system supporting farmers in procedurally and accurately detecting and controlling weeds in paddy fields. The farmers work directly with the system by using the PaddyGuru mobile application in which they take photos and upload the pictures of the weeds on their farms. Such images are then preprocessed, and a CNN model is applied to the images, which is trained on an extensive paddy weed image dataset labeled manually. The witnessed ability of the system to utilize an AI inference processing engine on the device allows the establishment of the weed type, its growth stage, and the factor contributing to the growth of the weeds, and minimizes the use of the internet as the system operates by following the edge computing protocol.

The results generated in the CNN model are displaying the results, giving the farmers insight into various information about the type of weed, type of growth stage, level of the growth, factors that cause the growth, and what kind of treatment or preventative action should be taken. The cloud synchronization continuously updates the model and provides enhancement on the data set to provide more effective detection and accurate real-time data to the farmers in order to better manage the paddy fields.

3.2.1 Software Solution

Technologies

- Programming Languages: Python (model training, CV processing), JavaScript (React Native/Node.js for mobile/web)

- Frameworks & Libraries: TensorFlow, PyTorch (deep learning), OpenCV, Pillow (image preprocessing)

- Databases: MongoDB, Firebase (store images, predictions)

- Cloud Services: AWS, Microsoft Azure (training, scalability, deployment)

- Version Control: GitHub, GitLab (source code management)

Techniques

- Image preprocessing (resizing, cropping, color normalization, augmentation)
- Transfer Learning (Using pre-trained CNNs)
- Model optimization for mobile(TensorFlow Lite)
- Real time inference on edge devices.

Algorithms

- Convolutional Neural Networks (CNN) – Weed detection
- Recurrent Neural Networks (RNN) – Growth stage identification
- Object Detection Algorithms – YOLO, Faster R-CNN,

Architecture

- AI Model Architecture: CNN-based image classification
- System Architecture: Mobile App (React Native), Backend (Node.js+ MongoDB), AI Model Service (Python + TensorFlow/ PyTorch)+ Cloud integration (AWS/Azure

- **Requirement gathering**

Data gathering

Our team is excited to visit the Rice Research and Development Institute (RRDI) in order to gather information. Experts in the institute keep records which pertain to rice farming, pest management, soil management and analysis of yield. At present, the majority of information is collected manually and we also hope to get some of the data in soft copy format. Moreover, we would like to obtain some more information through personal communication with researchers and observing the experiments they conduct.

Conducting a survey

We are planning to conduct a survey by distributing a questionnaire to gain an understanding of farmers' knowledge and practices regarding rice cultivation. The survey involved the use of modern methods of farming, awareness of the pest and disease controls methods, and the embracement of sustainable farming. This, we hope, will enable us to know the difficulties that farmers have and the expectations that they have on how things can be done better.

- **Design (system and software design documents)**

Sequence Diagram

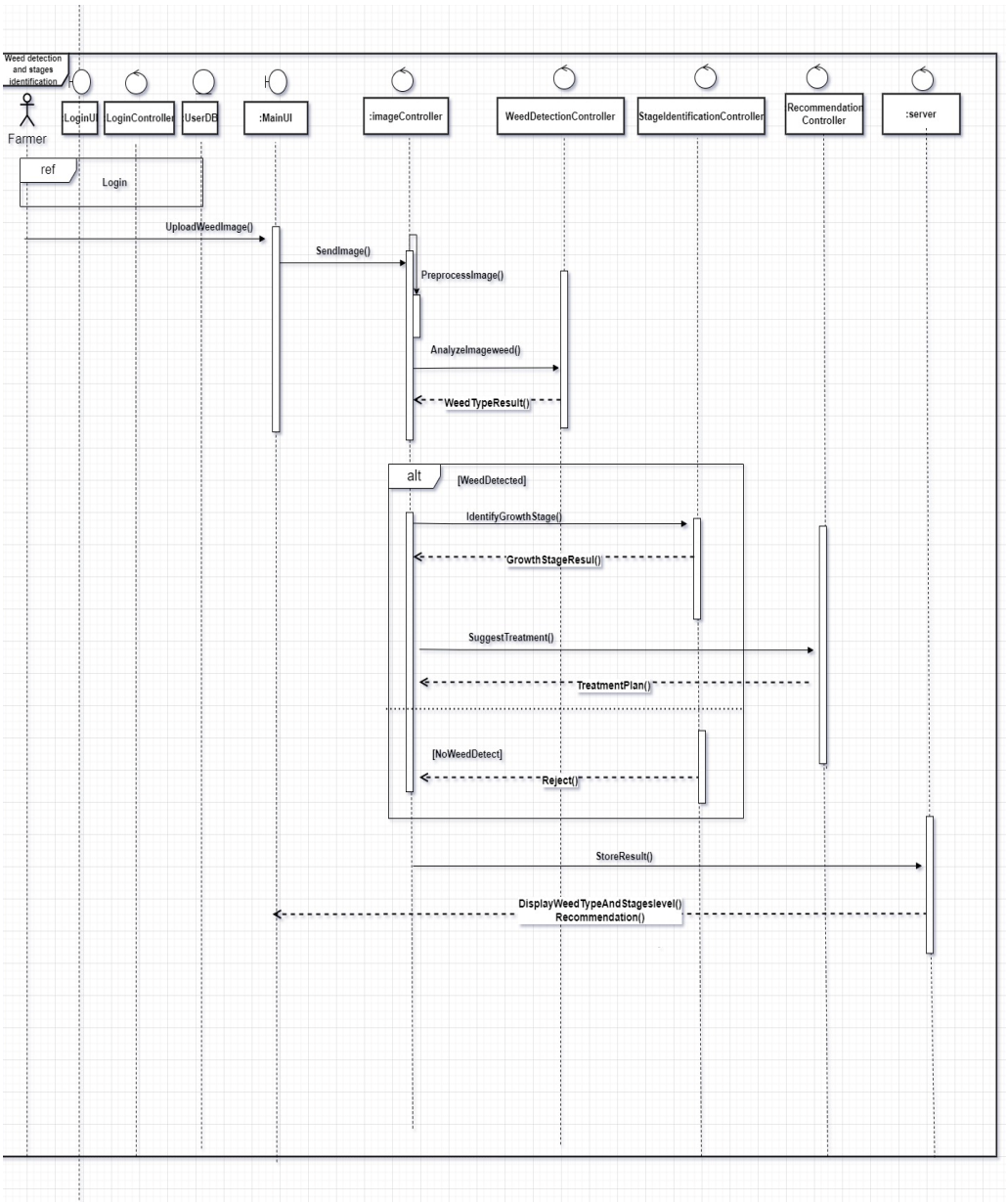

Figure 3.3:Sequence diagram of Weed detection identification and Stages Identification component

Use Case Diagram

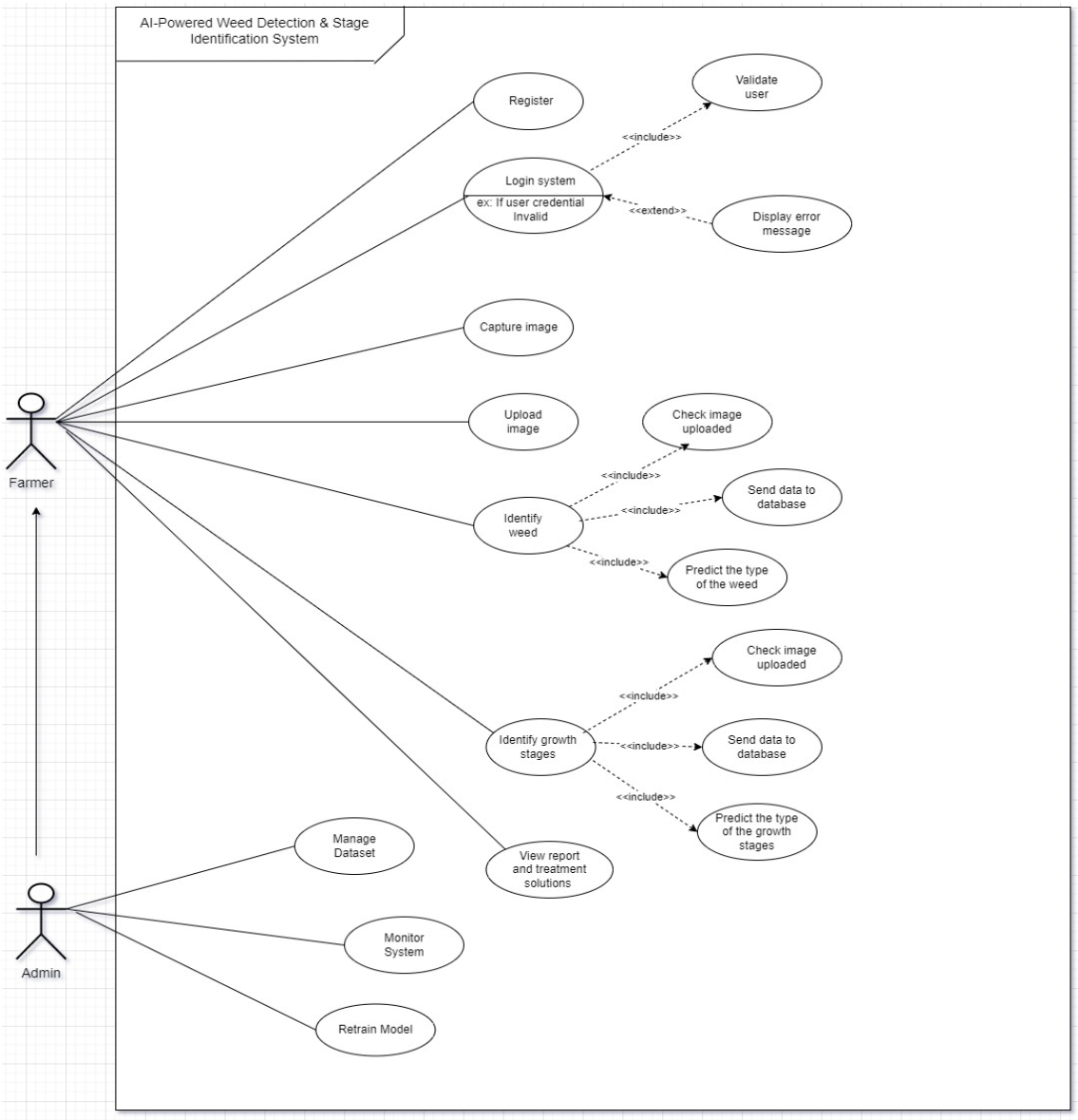

Figure 3.4:Use case diagram of Weed detection and stages identification

3.2.2 Commercialization

The introduction of new technology, the novel products of advanced computer vision, growth stage analysis, and locally-specific treatment suggestions, the brand-new AI-powered Weed Detection and Growth Stage Identification System can become the revolutionary new way of managing the process of paddy cultivation in Sri Lanka. This novel technology solves the major problems experienced by the paddy farmers in Sri Lanka and at the same time offers greater support in management of the weeds, cost reduction, and sustainable agriculture.

Commercialization Strategy:

Collaboration with Agricultural Organizations: Collaboration with the Department of Agriculture, provincial agricultural organizations, agribusiness and research institutions in Sri Lanka can be used to validate the system and report to/reach farmers, combine herbicide databases, and improve and update the system continuously.

Precision Agriculture Insights: The system provides insightful analytics on weed patterns, treatment efficacy, and correlate these data with yields; and it uses predictive models, to predict when the weed will emerge based on environment and seasonal data. It also gives cost-benefit analyses tabulating savings in herbicides and yield enhancement at an individual farm and regional levels, stable farming indicators in the reduction in environmental impact through more efficient use of chemicals.

Continuous Improvement: Pledge to continuously improve the system of weed detection and stage identification by training our AI models on new field sight data and using images shared by users of the product. Farmer feedback will be used to analyze misclassification and be used to improve the accuracy of the model across a variety of environments. Future developments will add or remove any changes to computer vision and deep learning (i.e., enhanced object detection algorithms) to maximize precision and achieve real-time performance. Also, the dynamic of

localized dataset hillock species and herbicide knowledge bases will be integrated to the system to make it updated with the emerging type of farming methods thus ensuring the system remains a valid and authoritative decision-aid tool to the farmers.

Targeted users: The targeted users of the system are paddy farmers, agricultural extension officers and suppliers of agrochemicals. It also facilitates public service agriculture departments, research organizations, and NGO allied to precision farming and sustainable agriculture.

Accessibility: The system will be accessible via a mobile application whereby the farmer will be able to capture field images and get a real-time result on weed detection and stage identification wherever farmer is. There will also be offline functionality so that we connect with the rural communities that have little internet connectivity.

Marketing Approach: Marketing efforts will be directed toward paddy farmers, agricultural cooperatives, and government agricultural agencies through industry-specific channels such as Agri exhibitions, farmer training programs, and local extension services. To expand outreach, social media campaigns, community awareness workshops, and partnerships with agrochemical suppliers will be employed, ensuring the solution reaches both small-scale and large-scale farmers.

3.2.3 Future Scope

The system has the capability of increasing to incorporate different types of crops other than paddy, which makes it flexible to various farming societies. The more complex models like Mask R-CNN could be incorporated to deliver specific detail of weed segmentation to perform analyses based on area covered, whilst drones and IoT capacity could be used to deliver large-scale coverage and real-time alerts. Future enhancements can also be done in the form of predictive analytics, whereby the climatic and soil patterns are studied to predict the patterns of the weed

evolution, and farmers can act beforehand by taking steps to prevent it. The control methods that the herbicide recommendation engine provides can be also regularly updated with locally-oriented, environmentally-friendly, and affordable solutions and maintained to keep the solution contemporary, and in relevance with agricultural demands.

4 PROJECT REQUIREMENTS

4.1 Functional requirements

- Image Capture and Processing
- Weed detection and classification
- Treatment and solutions
- User Interface (multi language) and Accessibility
- User feedback and Rating system

4.2 Non-functional requirements

- Performance and Efficiency
- Accuracy and Reliability
- Usability and User Experience
- Reliability and Availability
- Security and Privacy
- Compatibility and Integration
- Scalability and Maintainability

4.3 System requirements

- Visual Studio Code (VS Code) – To implement and manage the codebase.
- Python – For developing AI/ML models.
- TensorFlow / PyTorch – To train and deploy weed detection and growth stage identification models.
- OpenCV – For image preprocessing and feature extraction.
- Flask / FastAPI – To create an Application Programming Interface (API) for integrating the AI model with the system.
- React.js – To design and implement the frontend user interface.
- MongoDB / MySQL – To store datasets, model outputs, and user information.

- Postman – To test Application Programming Interface (API) endpoints.

4.4 User Requirements

Farmers:

The application will help farmers to identify the presence of weeds and see the growth stage of weeds using images of fields uploaded. The system will process the image automatically and point out the weeds occurrence. It will also determine the stage of crop growth, so that farmers can make appropriate and timely decisions like using herbicides or fertilizers at the right stage.

Agricultural Researchers:

The application will assist the researchers to collect the data regarding the patterns of the weed spread and weed growth. They can utilize this by uploading pictures of various regions to amass and evaluate data of the spread of weed and stages of growth. Such outputs will facilitate study of crop management techniques, weed control techniques, and increase in yield.

4.5 Wireframes

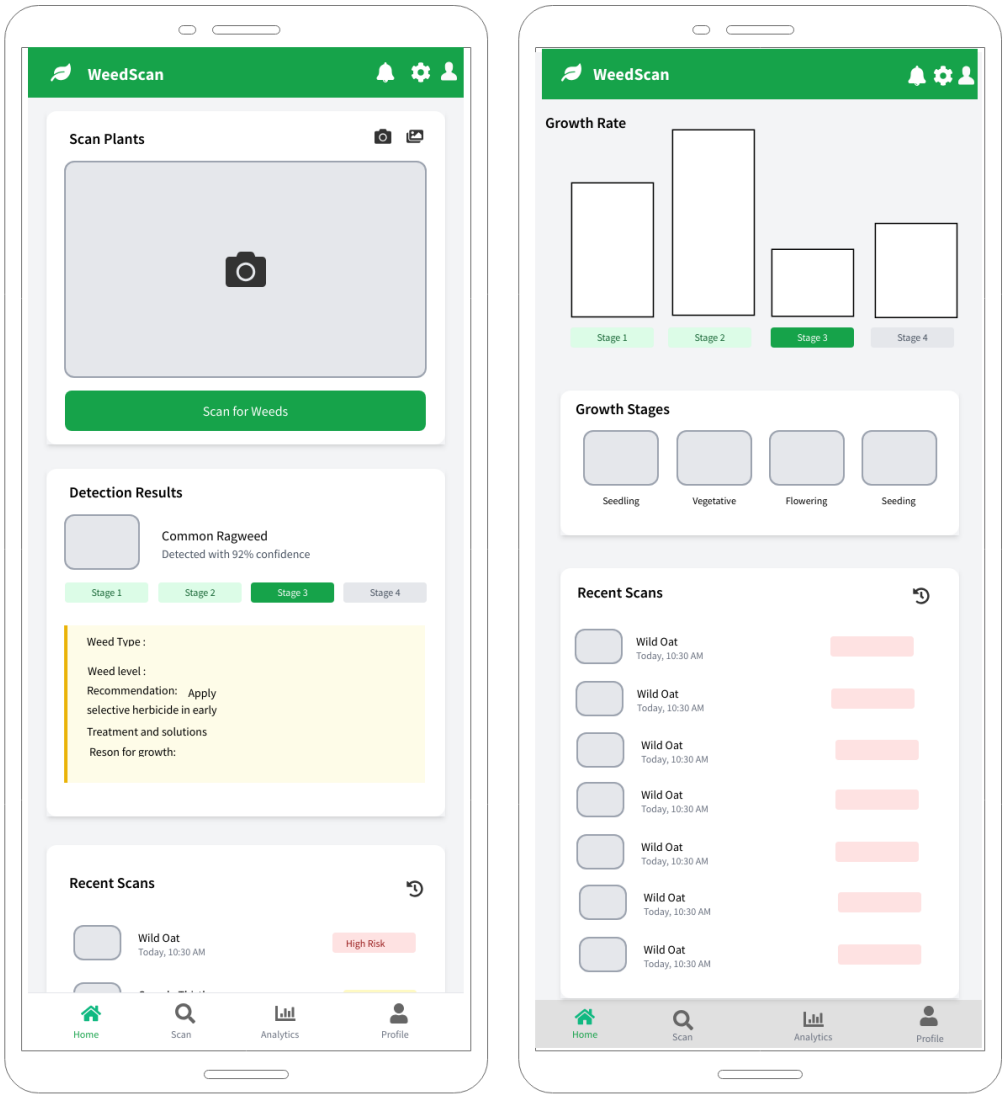

Figure 4.1:Wireframes of the process

5 GANTT CHART

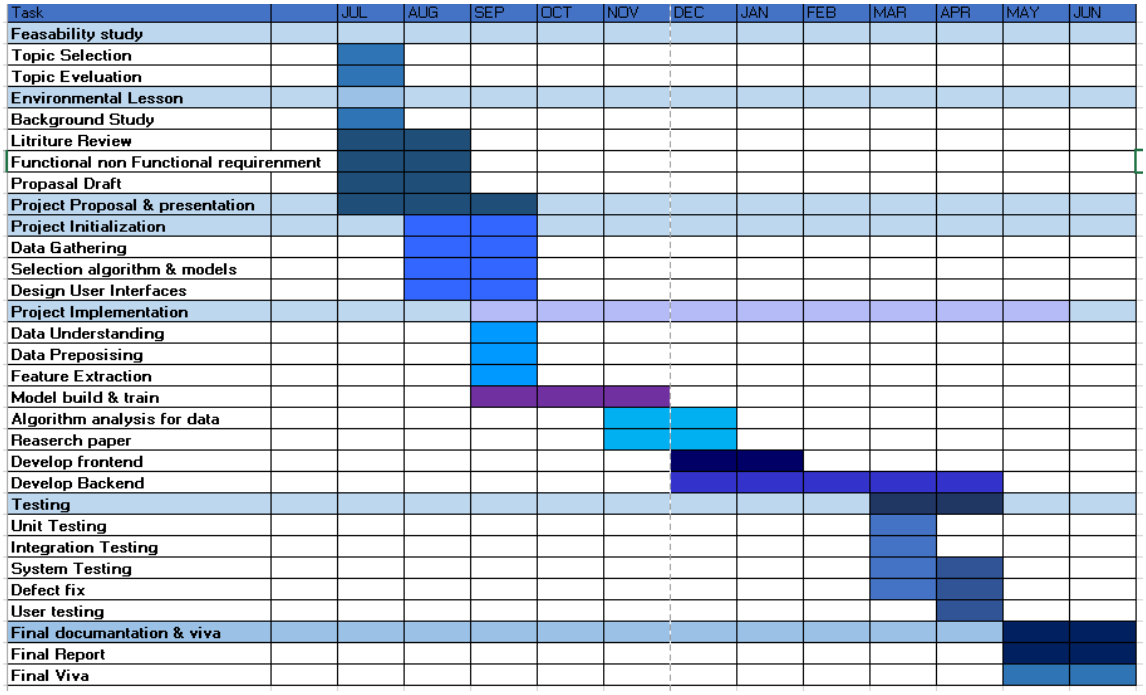

Figure 5.1:: Gantt Chart

6 WORK BREAKDOWN STRUCTURE

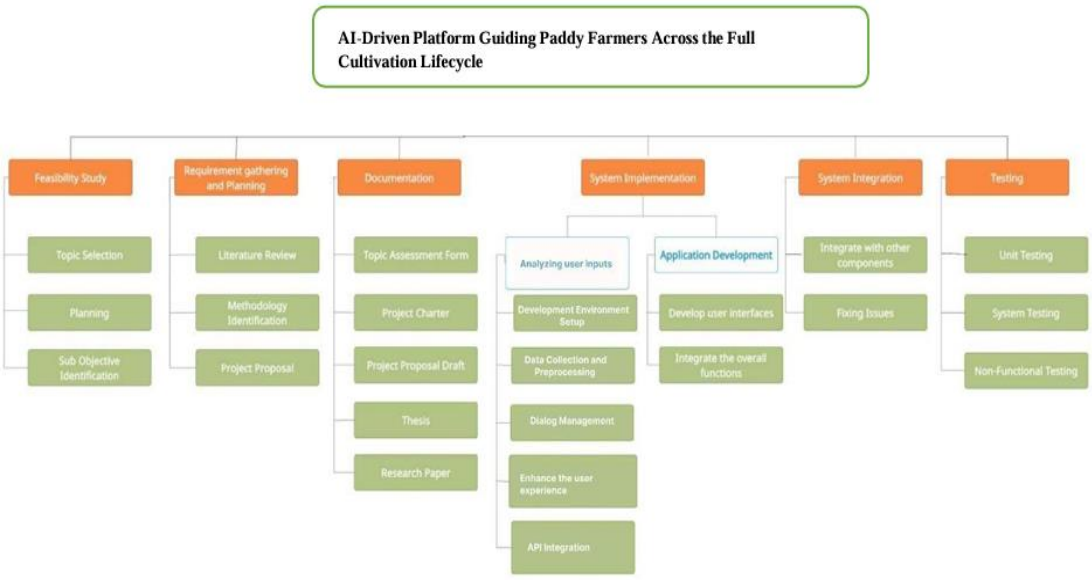

Figure 6.1:: Work Breakdown Chart of AI-Driven Platform Guiding paddy Farmers Acrooss the Full Cultivation Lifecycle

7 DESCRIPTION OF PERSONAL AND FACILITIES

7.1 Personal Qualifications and Expertise

Technical Expertise

Familiar with Python programming, with experience in popular machine-learning frameworks such as TensorFlow and PyTorch. Technical skills are knowledge of computer vision libraries like OpenCV which allow to combine advanced image processing algorithms into real-world applications. The hands-on experience of cross-platform mobile application development with the help of React Native can help create convenient solutions that are available to farmers and other stakeholders. Database design and management experience (SQL and MongoDB) will support storage and retrieval of agricultural data on a large scale. As well as, experienced in designing RESTful Application Programming Interface (API) and collaborating with cloud computing vendors to create high-performance, scalable, and secure AI-based agricultural solutions.

Agricultural Distribution Knowledge

Strong background knowledge on agriculture practices has been gained through widespread exposure to agricultural activities in Sri Lanka, especially in paddy cultivation. This experience gives outstanding insights into local field conditions and the obstacles farmers encounter during the different crop seasons. Knowledge has been gained about the main types of weeds common in Sri Lankan fields, about traditional methods of weed control - although often ineffectual or unsustainable. Real-life experience also shows constraints experienced by farmers in terms of cost, availability of resources, and speed of technology adoption. Professional knowledge in the agricultural sector can be transferred to digital groceries and other developments in the field of artificial intelligence, and thus ensure that solutions are not only up to date but also adapted to the needs of the sector and its culture.

7.2 Available Facilities and Resources

The use of SLIIT will be beneficial in the provision of various computing services that are necessary towards the successful execution of this research. These comprise access to powerful computing resources in training models, semblance environments specifically customized to the testing and even development of mobile applications. Alongside this, there are software licenses to professional development tools and applications, accessible reliable internet connection and cloud computing services that can be used to make bulk data processing and storage needs. The study will also be assisted by the existence of good working relationships with major players in the agricultural sector.

The Department of Agriculture Sri Lanka will offer expert advice and dataset validation and local agricultural extension services will assist in reaching the farmers and field testing. In addition, agricultural input suppliers will participate in building knowledge base related to herbicides and farmer organizations will be involved in collecting user feedbacks and confirming the feasibility of the system in actual conditions.

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8 BUDGET AND BUDGET JUSTIFICATION

The total proposed system budget is indicated in table 9.1 below.

Table 8.1:Budget Table

Requirements	**Cost(LKR)**
Development tools and licenses	10,000
Backend Hosting	12,000
Cloud computing	10,000
Field surveys & travel	15,000(5 trips)
Paper Publishing	15,000
Total	62,000

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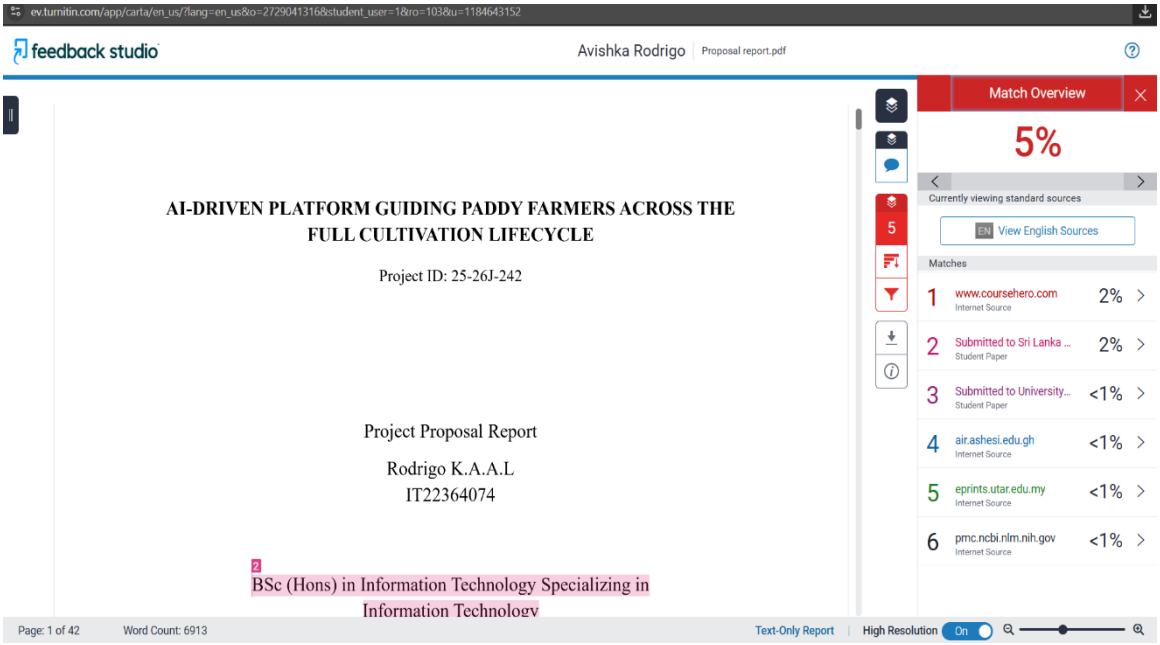
10 APPENDICES

10.1 Appendix: Plagiarism Report